Colloids and Surfaces A 604 (2020) 125291

Contents lists available at ScienceDirect

Colloids and Surfaces A

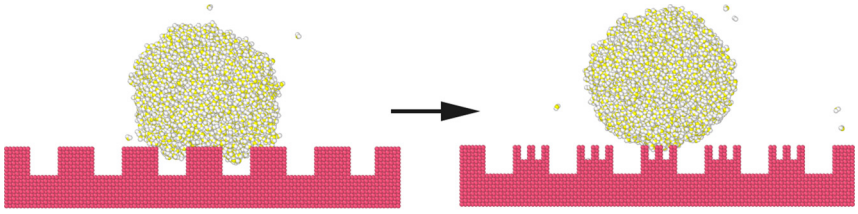
journal homepage: www.elsevier.com/locate/colsurfa

Wettability behavior of nanodroplets on copper surfaces with hierarchical nanostructures

Mingya Zhang[a], Lijun Ma[b], Qing Wang[a,*], Peng Hao[c], Xu Zheng[a]

[a] Institute of NanoEngineering, College of Civil Engineering and Architecture, Shandong University of Science and Technology, Qingdao 266590, China
[b] College of Pipeline and Civil Engineering, China University of Petroleum, Qingdao 266580, China
[c] College of Mining and Safety Engineering, Shandong University of Science and Technology, Qingdao 266590, China

GRAPHICAL ABSTRACT



Nonhierarchical structures are improved to hierarchical structures

Comparison of wettability behavior of different surface morphologies

ARTICLE INFO

Keywords:
Molecular dynamics simulation
Hierarchical gecko pillars
Hydrophobicity
Primary groove width
Surface characteristic energy

ABSTRACT

Wetting characteristics of nanodroplets on solid surfaces with nanostructures were investigated through molecular dynamics simulations. The effects of the primary groove width and the surface characteristic energy on the surface wettability with hierarchical structures and nonhierarchical structures were studied. Furthermore, the wettability on different surface morphologies was compared systematically. It is found that hierarchical gecko pillars tend to exhibit better hydrophobicity than nonhierarchical pillars. However, the increase of surface characteristic energy and primary groove width can weaken the advantage of hierarchical structures. Moreover, the contact angles gained from simulation were compared with the theoretical prediction values. It is expected that this study may have potential guiding significance in designing surface hydrophobicity.

1. Introduction

Micro- and nanostructures can be observed on the superhydrophobic surface of many animal skins and plants in nature, such as gecko feet [1,2], water strider [3,4], and lotus leaves [5–7]. Therefore, hydrophobicity property of a surface must be related to the surface topography as well as inherent properties of materials. The magnitude of contact angle is the metrics of surface wettability. The Wenzel [8] models and Cassie-Baxter [9] models are used for characterizing droplets on rough surfaces. The former asserts that the droplet completely fills the grooves of the solid surface, and the latter asserts that droplet sits above the pillar structures. The contact angles can be theoretically forecasted by the two models. In addition, wetting transition between two models is driven by many factors, such as surface pattern, electric field and so on. Zhang et al. [10] studied wetting transition on regularly nanostructured surfaces induced by an electric field. Ou et al. [11] developed a three-dimensional free energy to investigate the wetting transition for microstructured surfaces. However, the surface

* Corresponding author.
E-mail address: qwang@sdust.edu.cn (Q. Wang).

https://doi.org/10.1016/j.colsurfa.2020.125291
Received 14 May 2020; Received in revised form 21 June 2020; Accepted 8 July 2020
Available online 13 July 2020
0927-7757/ © 2020 Elsevier B.V. All rights reserved.

pattern of the material has a more important effect on the wetting transition.

Many researchers have carried out studies on building complex surface structures in recent years. Yoshimitsu et al. [12] investigated hydrophobicity of droplets on various hydrophobic pillarlike and groove structures. Oner and McCarthy [13] investigated hydrophobicity of rough surfaces patterned with various-shaped posts prepared by dry etching. Additionally, molecular dynamics simulations have been applied to investigate wettability at the nanoscale [14–16]. Yaghoubi et al. [17] studied the effect of surface roughness on wettability of nanodroplets. Hosseini et al. [18] investigated the effect of different surface morphologies and nanostructures on hydrophobicity by molecular dynamics, but they just considered a kind of surface characteristic energy. Sun et al. [19] studied the hydrophobicity on wavy and orthogonal textured surfaces. These studies have mainly investigated wetting behavior of nonhierarchical surfaces in both experiments and simulations. However, according to studies on natural hydrophobic phenomena such as gecko feet and lotus leaves, it is considered that the hierarchical surface is more hydrophobic than the nonhierarchical surface.

Therefore, hierarchical structural surfaces are also fabricated and investigated by experimental scheme. For example, Xu et al. [20–22] fabricated superhydrophobic surface with dendrite-like hierarchical structures on different substrates by chemical deposition and modification. Wang et al. [23,24] fabricated amphiphobic surfaces with hierarchical structures using a mixed modified solution of stearic acid and perfluorooctanoic acid. In addition, superhydrophobic surfaces with dendritic structures were fabricated on different substrates through vscuum evaporation, chemical replacement reaction, and heating [25–27]. Wang et al. [28] successfully simulated the superhydrophobic structures of the lotus leaf on bamboo surfaces by transferring micro/nano structures using soft lithography. It is found that droplets can easily slide down from the surfaces of such bamboo, exhibiting superhydrophobic and low adhesion properties. Liu et al. [29] successfully fabricated the superhydrophobic resultant textiles with the multi-scale structure of the duck feather. Park et al. [30] fabricated an artificial rose petal composed of hierarchical micro- and nanostructures on a polymethyl methacrylate surface, thereby showing good superhydrophobic performance. Although many researchers have conducted experimental studies on hierarchical surfaces, there are only limited studies on hierarchical surfaces at the nanoscale.

In this paper, how hierarchical structures affect hydrophobicity at the nanoscale was further investigated. Be inspired by the structure of the gecko feet, hierarchical column models similar to that were constructed. The wettability of hierarchical gecko pillars surfaces was compared with those of nonhierarchical pillars surfaces under different primary groove widths and surface characteristic energies using molecular dynamics simulations. And the simulated results are evaluated with the contact angles and water droplet shapes. This study is expected to provide significant insight towards the improvement of surface wetting characteristics.

2. Computational methods

2.1. Simulation details

To investigate the droplet behavior on the nonhierarchical pillar surface and hierarchical gecko pillar surface, the total number of water molecules that was initially set in a spherical shape was 6022 as illustrated in Fig. 1. Water molecules were modeled by three-site SPC/E model. The fix shake command was adopted to maintain the rigidity of the two O-H bonds and the H − O − H angle [31]. The electric charge of the oxygen atom and hydrogen atom are −0.8476 e and + 0.4238 e, respectively. The bond length between hydrogen atom and oxygen atom is 1 Å, and the bond angle between the hydrogen atoms is 109.47°. The solid substrate was made up of copper atoms whose lattice constant

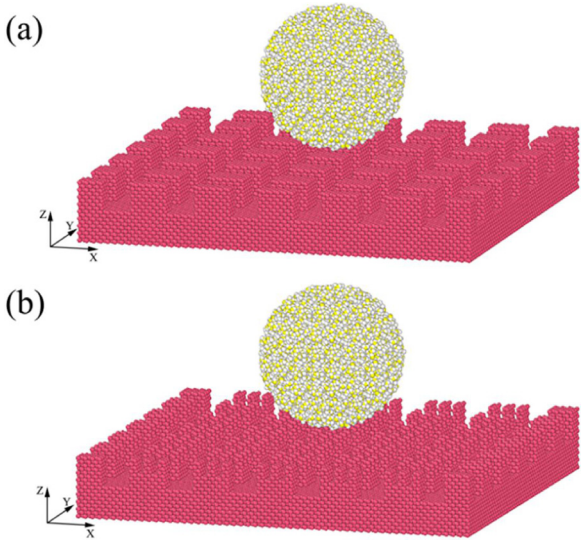


Fig. 1. Initial model of the simulation system with different surface morphologies: (a) nonhierarchical pillars (b) hierarchical gecko pillars.

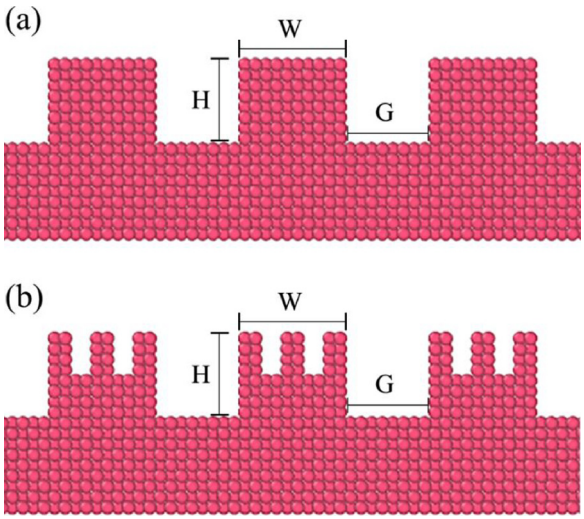


Fig. 2. Surface morphologies of the nonhierarchical pillars and hierarchical gecko pillars: (a) side view of nonhierarchical pillars (b) side view of hierarchical gecko pillars.

is 3.615 Å. And z-axis direction of solid surface is in contact with water molecules. The dimensions of the simulation box were 200 × 200 × 105 Å^3 in x, y and z direction, respectively. The surface morphologies of the columns are shown in Fig. 2. The hierarchical gecko pillars were composed of primary rectangular pillars patterned with some protuberances on their tops, whereas nonhierarchical pillars only consist of primary rectangular pillars. The pillar height H and width W for the primary rectangular pillars were 12.78 Å and 16.366 Å, respectively. The primary groove width G was changed from 3.615 to 16.366 Å.

The molecular dynamics approach was applied to simulate wetting behaviors which happen between water molecules and solid atoms. Simulations were performed with LAMMPS [32] in NVT ensemble at 298 K. Potential function refers to the interactions between molecules. The Lennard-Jones potential was used to calculate van der Waals force, defined as:

$$U_{LJ}(r) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \quad (1)$$

The electric potential was employed to calculate Coulombic force,

described as:

$$U_e(r) = \frac{q_i q_j}{Dr} \tag{2}$$

Where ε_{ij} is the characteristic energy, σ_{ij} is the characteristic length, r_{ij} is the distance between atoms i and j, D is the dielectric constant, and q_i and q_j are the partial atomic charges of atoms i and j, respectively. The LJ parameters for the oxygen-oxygen interactions are 0.1553 kcal/mol (ε_{oo}) and 0.3166 nm (σ_{oo}). Besides, ε_{os} and σ_{os} can be gained through the Lorentz-Bertholet mixing rule:

$$\varepsilon_{os} = \sqrt{\varepsilon_{oo} \times \varepsilon_s} \tag{3}$$

$$\sigma_{os} = \frac{\sigma_{oo} + \sigma_s}{2} \tag{4}$$

where subscript o and s represent oxygen atoms and solid atoms, respectively.

A periodic boundary condition was imposed in x and y directions. The cutoff radius of 10 Å and 12 Å were employed in Lennard-Jones and electrostatic interactions, respectively. The Particle-Particle Particle-Mesh K-space technique was adapted to calculate the long-range electrostatic interactions. The systems were run 700 ps to reach equilibrium and then run additional 300 ps to obtain accurate data. Models which consist of solid atoms and water molecules were built through Material Studio platform. The open visualization tool (OVITO) was applied for visual simulation results [33].

2.2. Contact angle measurement

Fig. 3 shows a schematic representation of contact angle measurement using constant density fitting curve method [34]. To calculate the contact angles of water droplets, first, the density profiles of a droplet in xz and yz views were drawn and then a circle was fitted on the liquid-gas boundary. Finally, the contact angles of the water droplets in xz and

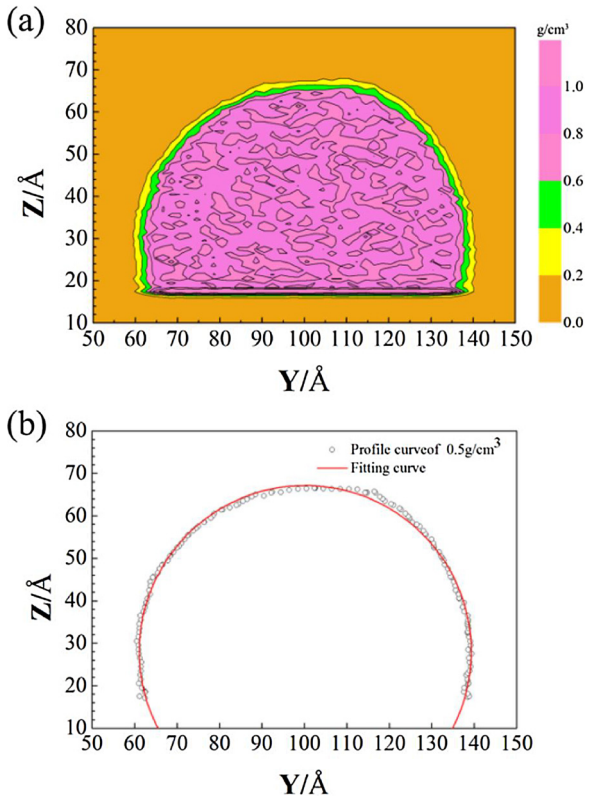


Fig. 3. Measurement of contact angle: (a) Density profile of water droplet (b) Fitting curve of water droplet.

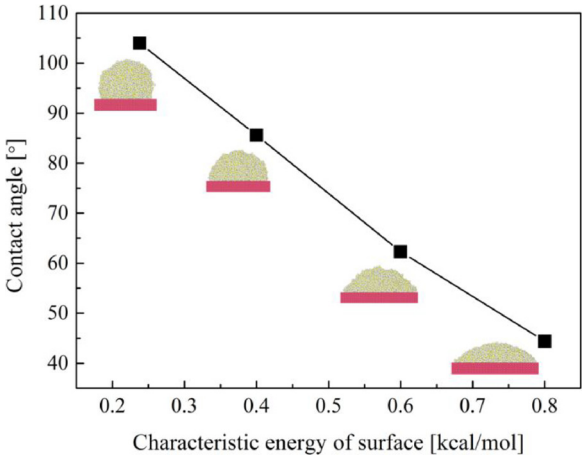


Fig. 4. Effects of different surface characteristic energies on contact angles of droplets on the smooth surface.

yz views were measured by calculating the slope of the tangent line to the circle at the triplet line. The profile curve of 0.5 g/cm^3 was considered as the liquid-gas boundary as illustrated in Fig. 3(a). The coordinates along the liquid-gas boundary were represented by small black circles and the fitting circle was represented by the red solid curve as shown in Fig. 3(b). The copper substrate is located in the blank area below the density profile, that is to say, approximately the position of Z < 1.6 nm.

3. Results and discussion

3.1. Water droplet in equilibrium on a smooth surface

To study the hydrophobicity of smooth surface, four different surface characteristic energies ε_S were constructed from 0.2379 kcal/mol to 0.8 kcal/mol. Fig. 4 exhibits the change in the water contact angle for different surface characteristic energies, as well as the corresponding water droplet snapshots on the smooth surface. As the increase of surface characteristic energy, the contact angle decreases continuously. This can be interpreted as follows: with large surface characteristic energy, the interactions between water molecules and solid atoms are much stronger, thus water molecules are dragged to the plain surface. Conversely, with small surface characteristic energy, the interactions inside of the water molecules are much stronger, so water molecules reunite with each other. It is found that our results are consistent with related studies [35,36]. In addition, the contact angle of 102° on a plain copper surface reported by Cheng et al. [37] was chosen as benchmark for this study. At the same time, the analog value of contact angle on the smooth copper surface ($\varepsilon_s = 0.2379 kcal/mol$) is 104.2°. Small deviation between simulation and experiment provides support for the credibility of this simulation.

3.2. Comparison of nonhierarchical surface with different surface characteristic energies and primary groove widths

In order to facilitate reading and understanding, subscript symbols were used to distinguish different cases: ε_S and G represent the surface characteristic energy and primary groove width, respectively, as displayed in Table 1. Fig. 5 shows the water droplet shapes on nonhierarchical pillar surfaces with four primary groove widths and three surface characteristic energies. The variation of the contact angles of nonhierarchical pillar surface according to the primary groove width for different surface characteristic energies is shown in Fig. 6.

For the same primary groove width, the lower the surface characteristic energy, the less the water molecules sink in interspaces among the pillars. The surface characteristic energy of ε_{S1} has weak interaction

Table 1
Subscript notations for the primary groove widths and surface characteristic energies.

Subscript	Primary groove width (G)	Surface characteristic energies (ε_S)
1	3.615 Å	0.2379 kcal/mol
2	7.229 Å	0.4 kcal/mol
3	12.78 Å	0.6 kcal/mol
4	16.366 Å	

on the droplets, therefore water droplets tend to form agglomeration and eventually stand on the surface of the pillars. In this condition, only the primary groove width of G_4 was in the Wenzel state. The contact angle increases as the primary groove width increases from G_1 to G_2 and then decreases as the primary groove width increases from G_2 to G_4. The peak contact angle appears at a primary groove width of G_2, which is attributed to the smaller gaps that stop water molecules from intruding into the spaces among the pillars. It is noteworthy to state that a portion of the water molecules penetrate to a position close to half height of the pillar at primary groove width of G_3. Furthermore, as primary groove width increases to G_4, the grooves among the columns are filled with water molecules, forming a transition from the Cassie-Baxter state to the Wenzel state. Wetting transition of nonhierarchical structures is a slow process accompanied with decreasing contact angle. The reason for latter two cases is that, as increasing primary groove width, the gaps which water molecules can penetrate increase, and the Van der Waals forces is much stronger as water molecules get closer to the bottom of groove. Therefore, in order to prevent the wetting transition of water droplets, the largest primary groove width should be limited.

In the case of ε_{S2} G_1, the water droplet aggregated into a sphere on the bottom of the groove. As the primary groove width increased to G_2, G_3 and G_4, however, the shape of water droplets changed into a semicircle. This means that the increasing primary groove width makes the nonhierarchical pillar structures more hydrophilic. For the surface characteristic energy of ε_{S3}, the contact angle increases as the primary groove width increases from G_1 to G_3 and then decreases as the primary groove width increases from G_3 to G_4. Some water molecules are attracted to diffuse inside the grooves in those cases whose surface characteristic energy is ε_{S3}.

3.3. Comparison of hierarchical surface with different characteristic energies and primary groove widths

Fig. 7 shows the water droplet shapes on hierarchical gecko pillar surfaces with four primary groove widths and three surface characteristic energies. Fig. 8 shows the contact angles of hierarchical gecko pillar surface at each primary groove width for different surface characteristic energies. For the same primary groove width, the contact angle is much smaller as surface characteristic energy increases. With

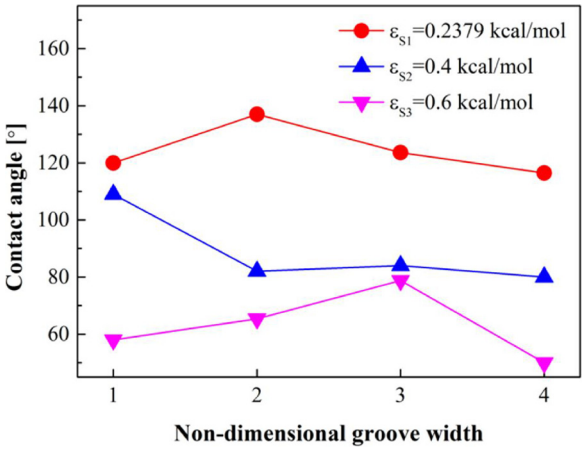


Fig. 6. Contact angles at each surface characteristic energy as the primary groove width changes for nonhierarchical pillar surfaces.

the surface characteristic energy of ε_{S1}, the simulated droplets for primary groove width of G_1-G_3 were in the Cassie-Baxter state and that for primary groove width of G_4 was in the Wenzel state. As primary groove width increases from G_1 to G_2, G_3 and G_4, the contact angles are 152.0°, 139.2°, 151.1°, and 115.9°, respectively. When the droplet is in the Cassie-Baxter state, increasing primary groove width has little effect on the water droplet of the hierarchical gecko pillar surface. When the primary groove width is W_3, water molecules stand perfectly on top of the hierarchical structure. This is because the van der Waals forces still can pull the water molecules to the surface. However, the van der Waals force becomes stronger as the primary groove widths increase to W_4. Therefore, an instant wetting transition happens on the surface of hierarchical structures, which is significantly different from the nonhierarchical structures. Meanwhile, the contact angle decreases abruptly as the droplet convert the Cassie-Baxter into Wenzel state.

For the surface characteristic energy of ε_{S2}, the contact angle decreases continuously as the primary groove width increases. This indicates the hierarchical gecko pillar structures become more hydrophilic by increasing the primary groove width. At the surface characteristic energy of ε_{S3}, as the primary groove width increases, the contact angle increases and then decreases. Water droplets on surface characteristic energy of ε_{S2} and ε_{S3} never reach the Cassie-Baxter state. In addition, small amount of water molecules flying are caused by the evaporation of water droplet at room temperature. This phenomenon was also reported in the relevant literature [38].

3.4. The comparison of different surface morphologies

Fig. 9 shows the contact angles of surfaces with nonhierarchical pillar and hierarchical gecko pillar for different surface characteristic energies. The square and the circle in Fig. 9 represent the contact angles

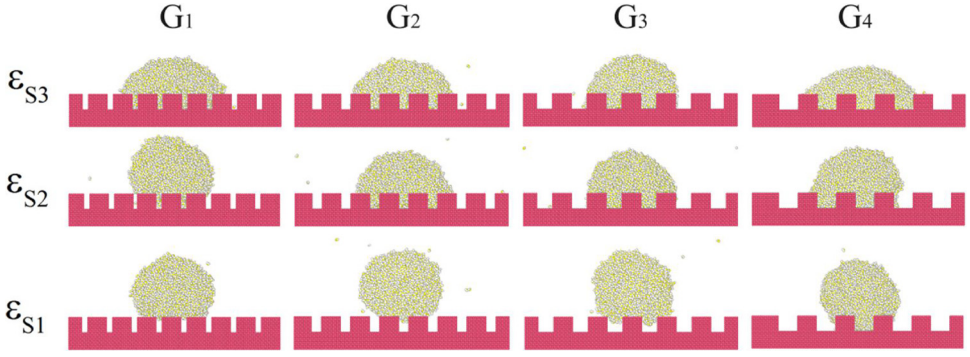


Fig. 5. Water droplet snapshots on nonhierarchical pillar surfaces with different primary groove widths and different surface characteristic energies.

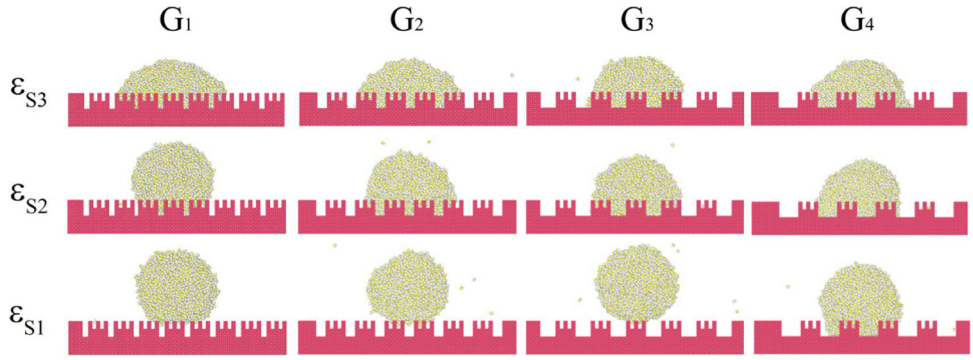


Fig. 7. Water droplet snapshots on hierarchical gecko pillar surfaces with different primary groove widths and different surface characteristic energies.

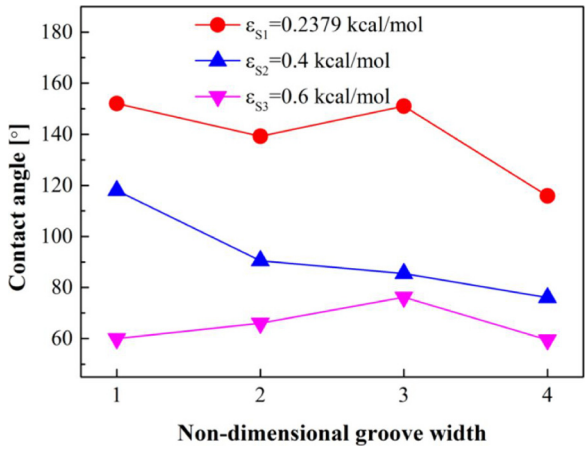


Fig. 8. Contact angles at each surface characteristic energy as the primary groove width changes for hierarchical gecko pillar surfaces.

of surfaces patterned with nonhierarchical and hierarchical structures, respectively. Correspondingly, the solid and hollow pentacles represent Wenzel and Cassie-Baxter theoretical contact angles, respectively. The solid blue lines and red dotted lines were employed to connect the symbols to better reflect the trend.

Fig. 9(a) shows the contact angles for the surfaces whose surface characteristic energy is ε_{S1}. The contact angles of hierarchical gecko pillars are greater than that of nonhierarchical pillars as primary groove width increases from G_1 to G_2 and G_3. This can be explained by the fact that the hierarchical gecko pillar is an improvement based on the nonhierarchical pillar, and some atoms have been deleted to form a shape similar to gecko feet. In this way, the gaps between the small pillars are so narrow that they can prevent the penetration of water molecules in a certain extent. On the other hand, the gaps between the small pillars can weaken the interactions between the water molecules and the solid atoms. Therefore, water droplets on the surface of hierarchical gecko pillar can even achieve a superhydrophobic state. This can explain the reason why the water droplets can stand on gecko feet in nature. To get more convincing results, however, more parameters need to be optimized to approximate natural behaviors. The contact angles calculated from simulation for hierarchical surface are in good agreement with predicated values by Cassie-Baxter model. This is because only a small amount of water molecules enter in spaces among the small pillars [39].

Fig. 9(b) shows the contact angles for the surfaces whose surface characteristic energy is ε_{S2}. Again, due to the hierarchical structures of the gecko pillars, they tend to exhibit better hydrophobicity than nonhierarchical pillar structures. As the primary groove width increases, however, the advantage of hierarchical structures is gradually weakened. When the primary groove width is G_4, the contact angle of hierarchical gecko column is smaller than that of nonhierarchical column. These contact angles closely approach the predicted values by Wenzel model except for the shortest primary groove width of G_1.

Fig. 9(c) shows the contact angles for the surfaces whose surface characteristic energy is ε_{S3}. Affected by the highest surface characteristic energy, the hierarchical structure loses its advantage. The contact angles of the hierarchical gecko pillars almost coincide with that of the nonhierarchical pillars. These contact angles obviously differ from the Wenzel model values, especially for shorter primary groove width. This indicated that Wenzel model cannot accurately predict the contact angles in these cases. This result also has been reported in other simulation studies [14,40].

In order to better distinguish the state of droplets in all cases, a map including the surface characteristic energy and the primary groove width for the surface with nonhierarchical pillars and hierarchical gecko pillars is shown in Fig. 10. The red circles and blue five-pointed stars represent the Wenzel state and the Cassie-Baxter state, respectively. Most of the cases were in the Wenzel state, and just a few were in the Cassie-Baxter state. These cases in Cassie-Baxter state occur at the lower surface characteristic energy and shorter primary groove width. Whether it is a hierarchical structure or a nonhierarchical structure, however, the Wenzel state can be obtained at all primary groove widths for the strong surface characteristic energy. Molecular dynamics simulations further confirm that superhydrophobic phenomena are caused by low surface characteristic energy and rough surface structures, which is a powerful complement to the experimental study on superhydrophobicity.

4. Conclusions

Wetting characteristics of water droplets on smooth and rough surfaces were investigated using molecular dynamics simulations. To investigate the surface wettability with hierarchical structures and nonhierarchical structures, four primary groove widths and three surface characteristic energies were constructed. The results indicated that the primary groove width has different impacts on the wettability according to the magnitude of surface characteristic energy. The results also indicated that the lower surface characteristic energy, more hydrophobic for the same geometry. The wetting transition of droplets verifies that the largest primary groove width should be restricted to stop droplets from wetting transition. When the droplet is in the Cassie-Baxter state, increasing primary groove width has small impact on the hydrophobicity of the hierarchical structures. Furthermore, the wettability of different surface morphologies was compared systematically. The results indicated that hierarchical gecko pillars are prone to demonstrate better hydrophobicity than nonhierarchical pillars. But the increase of surface characteristic energy and primary groove width can reduce the superiority of hierarchical structures. What's more, the contact angles calculated from simulation were compared with the theoretical prediction values. The results showed that the contact angles calculated from simulation are in good agreement with the Cassie-

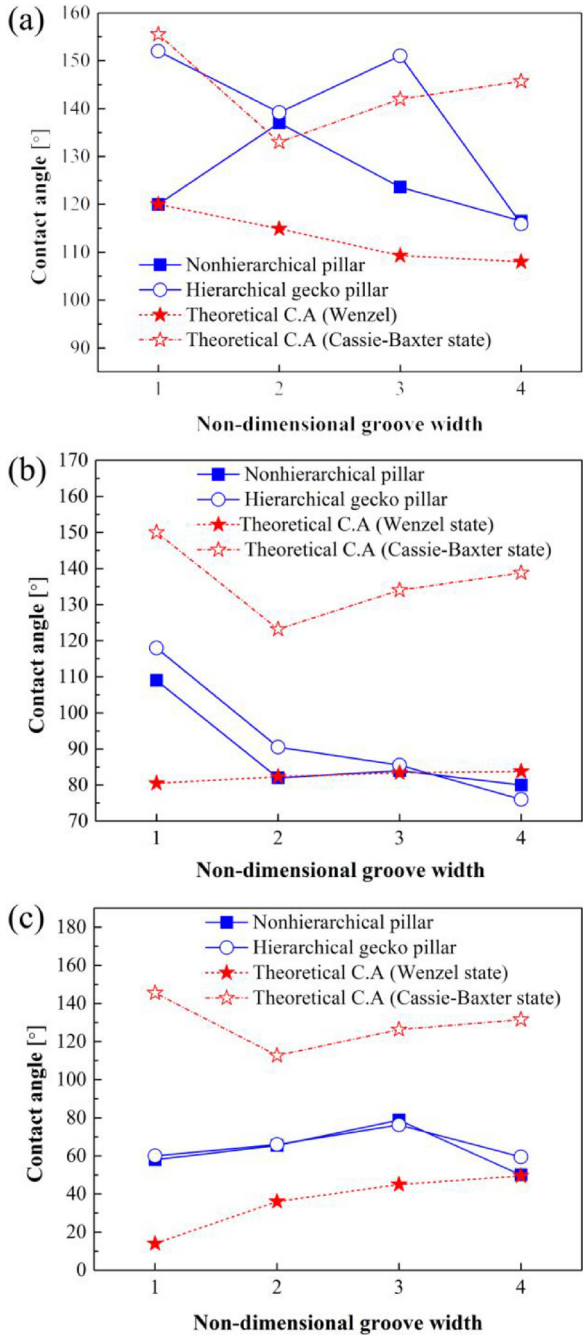


Fig. 9. Contact angles of surfaces with nonhierarchical pillar and hierarchical gecko pillar for different surface characteristic energies (a) 0.2379 kcal/mol (b) 0.4 kcal/mol (c) 0.6 kcal/mol.

Baxter model values for hierarchical gecko pillars surface. However, the contact angles calculated from simulation distinctly diverge from the Wenzel model values, especially for shorter primary groove width. This means that there are limitations at the nanoscale for Wenzel theory.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

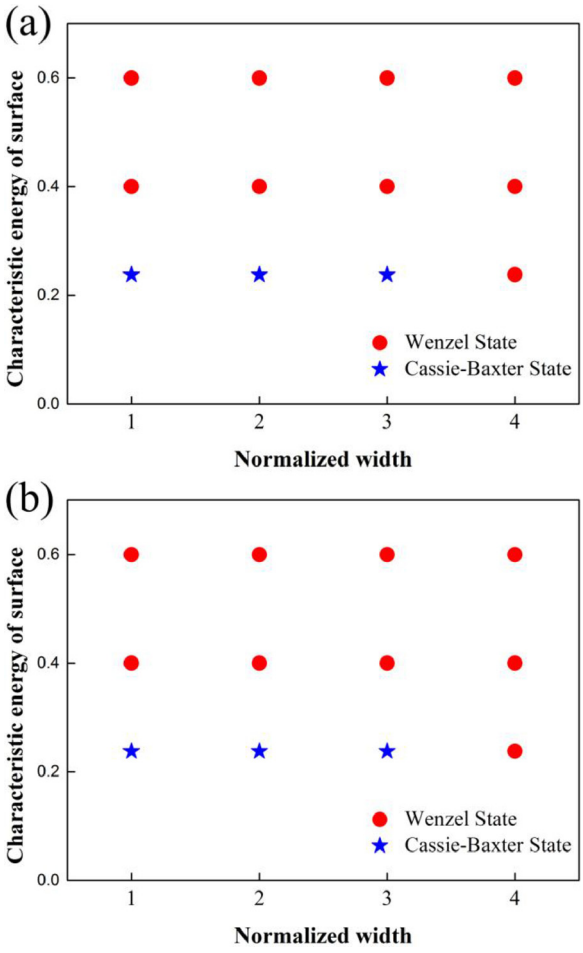


Fig. 10. Comparison of the Cassie-Baxter or Wenzel states of the water droplets on the surfaces with (a) nonhierarchical pillar (b) hierarchical gecko pillar.

Acknowledgments

This work was supported by the Taishan Scholar Project of Shandong Province (No. TSHW20130956) and the Natural Science Foundation of Shandong Province, China (No. ZR2017MA013).

References

[1] K. Autumn, N. Gravish, Gecko adhesion: evolutionary nanotechnology, Philos. Trans. Math. Phys. Eng. Sci. 366 (2008) 1575–1590.
[2] K.S. Liu, J.X. Du, J.T. Wu, L. Jiang, Superhydrophobic gecko feet with high adhesive forces towards water and their bio-inspired materials, Nanoscale 4 (2012) 768–772.
[3] X.Q. Feng, X.F. Gao, Z.N. Wu, L. Jiang, Q.S. Zheng, Superior water repellency of water strider legs with hierarchical structures: experiments and analysis, Langmuir 23 (2007) 4892–4896.
[4] Y.W. Su, B.H. Ji, Y.G. Huang, K.C. Hwang, Nature's design of hierarchical superhydrophobic surfaces of a water strider for low adhesion and low-energy dissipation, Langmuir 26 (2010) 18926–18937.
[5] T. Darmanin, F. Guittard, Superhydrophobic and superoleophobic properties in nature, Mater. Today 18 (2015) 273–285.
[6] S.H. Hsua, K. Woana, W.G. Sigmund, Biologically inspired hairy structures for superhydrophobicity, Mat. Sci. Eng. R 72 (2011) 189–201.
[7] S.H. Li, J.Y. Huang, Z. Chen, G.Q. Chen, Y.K. Lai, A review on special wettability textiles: theoretical models, fabrication technologies and multifunctional applications, J. Mater. Chem. A 5 (2017) 31–55.
[8] R.N. Wenzel, Resistance of solid surfaces to wetting by water, Ind. Eng. Chem. 28 (1936) 988–994.
[9] A.B.D. Cassie, S. Baxter, Wettability of porous surfaces, Trans. Faraday Soc. 40 (1944) 546–551.
[10] B.X. Zhang, S.L. Wang, X.D. Wang, Wetting transition from the Cassie-Baxter state to the Wenzel state on regularly nanostructured surfaces induced by an electric field, Langmuir 35 (2019) 662–670.
[11] J.F. Ou, G.P. Fang, W. Li, A. Amirfazli, Wetting transition on textured surfaces: a thermodynamic approach, J. Phys. Chem. C 123 (2019) 23976–23986.

[12] Z. Yoshimitsu, A. Nakajima, T. Watanab, K. Hashimoto, Effects of surface structure on the hydrophobicity and sliding behavior of water droplets, Langmuir 18 (2002) 5818–5822.
[13] D. Oner, T.J. McCarthy, Ultrahydrophobic surfaces. Effects of topography length scales on wettability, Langmuir 16 (2000) 7777–7782.
[14] L. Chen, S.Y. Wang, X. Xiang, W.Q. Tao, Mechanism of surface nanostructure changing wettability: a molecular dynamics simulation, Comput. Mater. Sci. 171 (2020) 109223.
[15] T. Koishi, K. Yasuoka, S. Fujikawa, T. Ebisuzaki, X.C. Zeng, Coexistence and transition between Cassie and Wenzel state on pillared hydrophobic surface, Proc. Natl. Acad. Sci. U. S. A. 106 (2009) 8435–8440.
[16] J.X. Chen, W.Y. Chen, Y.J. Xie, Z.G. Wang, J.B. Qin, Wettability behavior of water droplet on organic-polluted fused quartz surfaces of pillar-type nanostructures applying molecular dynamics simulation, Appl. Surf. Sci. 396 (2017) 1058–1066.
[17] H. Yaghoubi, M. Foroutan, Molecular investigation of wettability of rough surfaces by molecular dynamics simulation, Phys. Chem. Chem. Phys. 20 (2018) 1–36.
[18] S. Hosseini, H. Savaloni, M.G. Shahraki, Influence of surface morphology and nanostructure on hydrophobicity: a molecular dynamics approach, Appl. Surf. Sci. 485 (2019) 536–546.
[19] K.W. Sun, M.S. Ambrosia, T.W. Kwon, M.Y. Ha, A hydrophobicity study on wavy and orthogonal textured surfaces, Comput. Fluids 140 (2016) 347–356.
[20] S.S. Xu, Q. Wang, N. Wang, X. Zheng, L.L. Lei, Environmentally-friendly fabrication of a recyclable oil-water separation material using copper mesh for immiscible oil/water mixtures, Colloids Surf. A 583 (2019) 124010.
[21] S.S. Xu, Q. Wang, N. Wang, X. Zheng, Fabrication of superhydrophobic green surfaces with good self-cleaning, chemical stability and anti-corrosion properties, J. Mater. Sci. 54 (2019) 13006–13016.
[22] S.S. Xu, Q. Wang, N. Wang, Eco-friendly fabrication of superhydrophobic surface with anti-corrosion by transferring dendrite-like structures to aluminum substrate, Colloids Surf. A 595 (2020) 124719.
[23] N. Wang, Q. Wang, S.S. Xu, X. Zheng, M.Y. Zhang, Facile fabrication of amphiphobic surfaces on copper substrates with a mixed modified solution, RSC Adv. 9 (2019) 17366–17372.
[24] N. Wang, Q. Wang, S.S. Xu, X. Zheng, Fabrication of multifunctional amphiphobic surfaces on copper substrates, Colloids Surf. A 577 (2019) 509–516.
[25] N. Wang, Q. Wang, S.S. Xu, X. Zheng, Eco-friendly and safe method of fabricating superhydrophobic surfaces on stainless steel substrates, J. Phys. Chem. C 123 (2019) 25738–25746.
[26] N. Wang, Q. Wang, S.S. Xu, X. Zheng, Mechanical stability of PDMS-based micro/nanotextured flexible superhydrophobic surfaces under external loading, ACS Appl. Mater. Interfaces 11 (2019) 48583–48593.
[27] N. Wang, Q. Wang, S.S. Xu, J.J. Luan, Flexible films with wrinkled micro-nano hierarchical structures having stable superhydrophobicity under external loading, J. Mater. Sci. 55 (2020) 9623–9637.
[28] F.P. Wang, S. Li, L. Wang, Fabrication of artificial super-hydrophobic lotus-leaf-like bamboo surfaces through soft lithography, Colloids Surf. A 513 (2017) 389–395.
[29] Y.Y. Liu, X.Q. Chen, J.H. Xin, Hydrophobic duck feathers and their simulation on textile substrates for water repellent treatment, Bioinspir. Biomim. 3 (2008) 046007.
[30] Y.M. Park, M. Gang, Y.H. Seo, B.H. Kim, Artificial petal surface based on hierarchical micro- and nanostructures, Thin Solid Films 520 (2011) 362–367.
[31] J.P. Ryckaert, G. Ciccotti, H.J.C. Berendsen, Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes, J. Comput. Phys. 23 (1977) 327–341.
[32] S. Plimpton, Fast parallel algorithms for short-range molecular dynamics, J. Comput. Phys. 117 (1995) 1–19.
[33] A. Stukowski, Visualization and analysis of atomistic simulation data with OVITO-the open visualization tool, Modell. Simul. Mater. Sci. Eng. 18 (2010) 015012.
[34] J. Zhang, M.K. Borg, K. Sefiane, J.M. Reese, Wetting and evaporation of salt-water nanodroplets: a molecular dynamics investigation, Phys. Rev. E 92 (2015) 052403.
[35] M.J. Yoo, M.S. Ambrosia, T.W. Kwon, J. Jang, M.Y. Ha, Wetting characteristics of a water droplet on solid surfaces with various pillar surface fractions under different conditions, J. Mech. Sci. Technol. 32 (2018) 1593–1600.
[36] J.A. Ko, M. Ambrosia, M.Y. Ha, A study of the wetting characteristics of a nano-sized water droplet on heterogeneous striped surfaces, Comput. Fluids 112 (2015) 19–34.
[37] Z.J. Cheng, M. Du, H. Lai, N.Q. Zhang, K.N. Sun, Super-hydrophobic copper surface with controlled adhesion prepared via ammonia corrosion, Chem. J. Chinese U. 34 (2013) 606–609.
[38] M.S. Ambrosia, J. Jang, M.Y. Ha, Static and dynamic hydrophobicity on a nano-sized groove/ridge surface, Comput. Fluids 114 (2015) 75–83.
[39] M.S. Ambrosia, M.Y. Ha, S. Balachandar, The effect of pillar surface fraction and pillar height on contact angles using molecular dynamics, Appl. Surf. Sci. 282 (2013) 211–216.
[40] M.S. Ambrosia, M.Y. Ha, A molecular dynamics study of Wenzel state water droplets on anisotropic surfaces, Comput. Fluids 163 (2018) 1–6.